ALEENA SAJI

Software Engineer
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PROFESSIONAL SUMMARY

B.Tech Computer Science and Engineering graduate and fresher Flutter Developer with strong Flutter and Dart mobile application development skills. Independently built **Double Bell** and released on Google Play, a Flutter and Firebase Android application, using Firebase Authentication, Cloud Firestore, Riverpod state management and the Google Play Console release workflow. Solid software engineering fundamentals in Java, Python, SQL, REST/API concepts, data structures and debugging, with supporting experience in full-stack web development, machine learning and generative AI applications.

TECHNICAL SKILLS

Mobile Development: Flutter, Dart, Android Development, Android Studio, Mobile Application Architecture, Responsive & Adaptive UI, UI Development, Animation

State Management & App Development: Riverpod, Provider, GoRouter, Shared Preferences, Lottie

Backend, APIs & Databases: Firebase, Firebase Authentication, Cloud Firestore, REST APIs, JSON, SQL, PostgreSQL, MongoDB, Node.js, Express.js

Programming Languages: Dart, Java, Python, JavaScript, C, C#

Software Engineering: Git, GitHub, Debugging, Testing, Data Structures, Problem Solving, Design Patterns, Google Play Console, Visual Studio Code

Web Development: HTML, CSS, Bootstrap, JavaScript, PHP, Full-Stack Web Development

AI / ML / Data: Machine Learning, Deep Learning, Computer Vision, Generative AI, Data Science, SVM, Random Forest, Streamlit

PROJECT EXPERIENCE

Double Bell – Flutter Android Application

Flutter, Dart, Riverpod, Firebase Authentication, Cloud Firestore, Google Sign-In, GoRouter, Git/GitHub

- Independently developed an end-to-end **Flutter/Dart Android application** for private bus travel, enabling schedule search by start location, destination, and arrival time; launched on Google Play.
- Integrated **Firebase Authentication, Google Sign-In, and Cloud Firestore** for user accounts, bus stop/schedule data, profile management, password changes, logout, and account deletion.
- Implemented **Riverpod** state management, **GoRouter** navigation, **Shared Preferences** persistence, responsive UI, **Lottie** animations, custom splash screen, **audioplayers**, and **URL launching**.
- Built time-based bus search with schedule display, **bus type information, incorrect-timing reporting, language selection**, and About, Privacy Policy, and Terms screens.
- Configured **Firebase production environment and Firestore security rules**, resolved application issues, and managed **Google Play** production release.

MammoSafe – Breast Cancer Detection Using Ultrasound, PET/CT and Histopathological Images

B.Tech Group Project | Python, Machine Learning, Deep Learning, Computer Vision, EfficientNet, SVM

- Developed the **histopathological image analysis module** as part of a 4-member AI-based breast cancer detection project using **Python**.
- Utilized **EfficientNet** for deep feature extraction from histopathological images and extracted discriminative features for accurate classification.
- Implemented **Support Vector Machine (SVM)** for classification and performed **image preprocessing, model training, and evaluation** for the histopathology component.
- Contributed to integrating histopathology predictions with **ultrasound and PET/CT models** using a **multi-modal fusion** approach to combine complementary medical imaging information.
- Collaborated with team members to integrate the machine learning models into a **Streamlit-based web application** supporting image upload, real-time prediction, and report generation.

AI-Powered Health & Nutrition Analysis App

Python, Streamlit, Google Gemini Vision API, Generative AI, PIL, python-dotenv

- Built a Streamlit application that uses the Gemini Vision API for multimodal analysis of food images, identifying food items and estimating calories, protein, carbohydrates and fats for each.
- Implemented image upload, custom user prompts and AI-generated nutritional reports; processed images with PIL, managed API keys with python-dotenv and organized code into reusable functions.

Multi-Language Invoice Extractor

Python, Streamlit, Google Gemini 1.5 Flash API, Generative AI, PIL, python-dotenv

- Developed a Streamlit application that uses Gemini 1.5 Flash to process invoice images (JPG, JPEG, PNG) and answer natural-language queries about their content.
- Built an interface with invoice upload, image preview and custom prompts; handled images with PIL and secured API credentials using environment variables.

EDUCATION

Bachelor of Technology (B.Tech) – Computer Science and Engineering **2022 – 2026**
University College of Engineering, Thodupuzha (APJ Abdul Kalam Technological University) | CGPA: 8.25

Higher Secondary Education
St. Mary's HSS, Arakkulam | Score: 96.33%

SSLC
Sacred Heart English Medium Higher Secondary School, Moolamattom | Result: Full A+

CERTIFICATIONS

- **Flutter & Dart – The Complete Guide** – Udemy (Academind / Maximilian Schwarzmüller)
- **The Complete Full-Stack Web Development Bootcamp** – Udemy (Dr. Angela Yu)
- **Artificial Intelligence using Python** – Cozmek Private Limited, Kothamangalam (recognized as one of the best students)
- **Python for Data Science** – NPTEL
- **Foundational C# with Microsoft** – freeCodeCamp
- **Brototype Campus Community** – Completion Certificate

LEADERSHIP & ACTIVITIES

- **Robotics Club, University College of Engineering, Thodupuzha** – Member for 4 years; Core Member for the final 1.5 years. Took part in technical discussions, shared ideas and volunteered for programs, events and games.
- **Python Classes for B.Tech Students** – Conducted Python classes, building communication, mentoring and technical knowledge; recognized with a reward for teaching.
- **Election Grievances Redressal Cell** – Member; handled and resolved election complaints.

LANGUAGES

English, Malayalam, Hindi, Russian

REFERENCE

Mrs. Rahumath Beevi – HoD, Department of Computer Science and Engineering, University College of Engineering, Thodupuzha | Contact: +919207557512